31/12/2020

Aakash
Medical | IIT-JEE | Foundations
(Divisions of Aakash Educational Services Limited)

***Regd. Office*:** Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 3

Answer Key

1.	(2)	37.	(3)	73.	(2)	109.	(2)	145.	(1)
2.	(2)	38.	(4)	74.	(3)	110.	(4)	146	(1)
3.	(2)	39.	(2)	75.	(2)	111.	(2)	147.	(3)
4.	(4)	40.	(2)	76.	(3)	112.	(2)	148.	(3)
5.	(1)	41.	(4)	77.	(4)	113.	(3)	149.	(2)
6.	(2)	42.	(2)	78.	(3)	114.	(3)	150.	(3)
7.	(1)	43.	(3)	79.	(2)	115.	(1)	151.	(3)
8.	(3)	44.	(4)	80.	(3)	116.	(2)	152.	(1)
9.	(3)	45.	(3)	81.	(4)	117.	(4)	153.	(3)
10.	(2)	46.	(3)	82.	(2)	118.	(3)	154.	(4)
11.	(4)	47.	(4)	83.	(2)	119.	(2)	155.	(3)
12.	(4)	48.	(3)	84.	(2)	120.	(3)	156.	(1)
13.	(3)	49.	(2)	85.	(4)	121.	(3)	157.	(1)
14.	(2)	50.	(4)	86.	(2)	122.	(1)	158.	(4)
15.	(4)	51.	(2)	87.	(4)	123.	(2)	159.	(4)
16.	(3)	52.	(1)	88.	(2)	124.	(4)	160.	(4)
17.	(2)	53.	(1)	89.	(2)	125.	(3)	161.	(1)
18.	(3)	54.	(2)	90.	(3)	126.	(1)	162.	(4)
19.	(2)	55.	(2)	91.	(2)	127.	(2)	163.	(4)
20.	(1)	56.	(2)	92.	(1)	128.	(1)	164.	(2)
21.	(4)	57.	(3)	93.	(4)	129.	(4)	165.	(2)
22.	(3)	58.	(2)	94.	(1)	130.	(4)	166.	(4)
23.	(3)	59.	(1)	95.	(3)	131.	(4)	167.	(4)
24.	(2)	60.	(4)	96.	(4)	132.	(3)	168.	(2)
25.	(2)	61.	(3)	97.	(2)	133.	(2)	169.	(2)
26.	(4)	62.	(2)	98.	(2)	134.	(3)	170.	(1)
27.	(1)	63.	(3)	99.	(3)	135.	(2)	171.	(2)
28.	(2)	64.	(2)	100.	(1)	136.	(4)	172.	(3)
29.	(2)	65.	(2)	101.	(4)	137.	(3)	173.	(4)
30.	(4)	66.	(3)	102.	(3)	138.	(1)	174.	(3)
31.	(1)	67.	(3)	103.	(3)	139.	(4)	175.	(2)
32.	(1)	68.	(1)	104.	(2)	140.	(1)	176.	(3)
33.	(3)	69.	(3)	105.	(1)	141.	(3)	177.	(4)
34.	(1)	70.	(4)	106.	(1)	142.	(4)	178.	(1)
35.	(4)	71.	(2)	107.	(3)	143.	(2)	179.	(1)
36.	(4)	72.	(3)	108.	(3)	144.	(3)	180.	(4)

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Test - 3

Hints and Solutions

PHYSICS

1. Answer (2)

$a_1 = \frac{F_1}{m}$

$a_1 = 4\ \text{m/s}^2$

$a_2 = \frac{F_2}{m}$

$a_2 = 2\ \text{m/s}^2$

$v_1 = v + a_1 t$

$v_1 = 4 \times 4 = 16\ \text{m/s}$

$v_2 = v + a_2 t$

$= 2 \times 4 = 8$

$\overrightarrow{v_{c.m.}} = \frac{m_1\vec{v_1} + m_2\vec{v_2}}{m_1 + m_2}$

$\overrightarrow{v_{c.m.}} = \frac{3 \times 16\hat{i} + 3 \times 8\hat{j}}{6}$

$\overrightarrow{v_{c.m.}} = (8\hat{i} + 4\hat{j})\ \text{m/s}$

2. Answer (2)

Moment of inertia of disc is minimum about a diameter.

3. Answer (2)

$a_R = \frac{g\sin\theta}{1 + \frac{K^2}{R^2}} = \frac{g\sin\theta}{1 + \frac{2}{3}} = \frac{3}{5}$

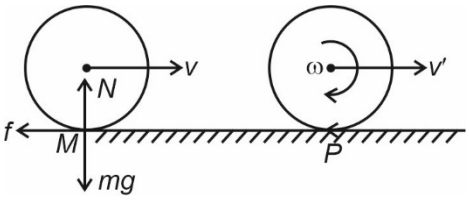
$a_S = g\sin\theta$

Hence, $\frac{a_R}{a_s} = \frac{3}{5}$

4. Answer (4)

Coefficient of friction between the object and inclined is same. Acceleration of each object will be same and equal to $a = g\sin\theta - \mu g\cos\theta$, therefore all object will take same time to reach the bottom.

5. Answer (1)



By conservation of angular momentum about the point of contact. (Torque about *P* is always zero.)

$mvR = I_P\omega$

$mvR = \left(\frac{mR^2}{2} + mR^2\right)\omega = \frac{3}{2}mR^2\omega$

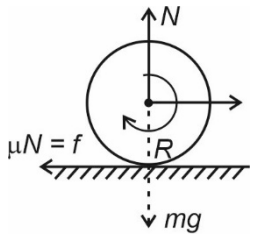
$\omega = \frac{2}{3}\frac{v}{R}\ \therefore\ v' = \omega R = \frac{2}{3} \times \frac{v}{R}R = \frac{2}{3}v$

$\Rightarrow\ v' = \frac{2v}{3}$

6. Answer (2)

$$\mu = \frac{\frac{K^2}{R^2}\tan\theta}{1+\frac{K^2}{R^2}} = \frac{\frac{1}{2}\tan 45^\circ}{1+\frac{1}{2}} = \frac{1}{3}$$

7. Answer (1)



For pure translation

$F - f = ma$

$F - \mu N = ma$ … (1)

For pure rotation

$f \times R = I\alpha$

$\mu NR = \frac{2mR^2}{5}\left(\frac{a}{R}\right)$ (Instant pure rolling)

$\frac{5\mu N}{2} = ma$ … (2)

$F - \mu N = \frac{5\mu N}{2}$

$F = \frac{7\mu N}{2}$ ($N = mg$)

$\mu = \frac{2F}{7\,mg}$

8. Answer (3)

$\omega = \omega_0 + \alpha t$

$\alpha = \frac{-40}{5}$

$\alpha = -8$ rad/s^2

$|\tau| = I\alpha$

$= 5 \times 10^{-2} \times 8$

$|\tau| = 40 \times 10^{-2} = 0.4$ Nm

9. Answer (3)

$W_{net} = \Delta K$

$400 = \frac{1}{2}I\left(\omega_2^2 - \omega_1^2\right)$

$400 = \frac{1}{2}I\,4\pi^2\left(f_2^2 - f_1^2\right) = 2\pi^2(5^2 - 3^2)I$

$I = \frac{400}{20(25-9)}$

$I = \frac{20}{16} = 1.25$ kg m^2

10. Answer (2)

By conservation of angular momentum

$\omega_1 I_1 = \omega_2 I_2 \Rightarrow 2I\omega = I\omega_2$

$\omega_2 = 2\omega$

11. Answer (4)

$$\frac{K_t}{K_r + K_t} = \frac{\frac{1}{2}MV^2}{\frac{1}{2}MV^2 + \frac{1}{2}I\omega^2}$$

$$= \frac{\frac{1}{2}MV^2}{\frac{1}{2}MV^2 + \frac{1}{2}MV^2} = \frac{1}{2}$$

12. Answer (4)

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ 1 & 1 & 2 \end{vmatrix}$$

$= \hat{i}(-2-1) - \hat{j}(6-1) + \hat{k}(3+1)$

$= -3\hat{i} - 5\hat{j} + 4\hat{k}$

$\left|\vec{A}\times\vec{B}\right| = \sqrt{(-3)^2 + (-5)^2 + 4^2} = 25\sqrt{2}\ m^2$

13. Answer (3)

When a mass is rotating in a plane about a fixed point. Its angular velocity is directed along a line perpendicular to plane of rotation.

14. Answer (2)

$\vec{\tau} = \vec{r}\times\vec{F}$

15. Answer (4)

By perpendicular axis theorem

$I = I_1 + I_2$ or $I = I_3 + I_4$

Also by symmetry

$I_2 = I_3$, $I_1 = I_4$

$\therefore$ $I = I_1 + I_3$ (All are correct)

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16. Answer (3)

By conservation of angular momentum

$I\omega + 2I\omega = 2I\omega_1$

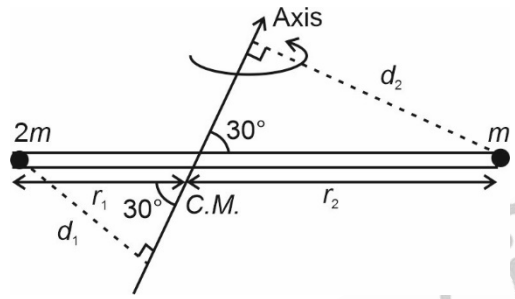
$\omega_1 = \left(\frac{3\omega}{2}\right)$

Loss in energy = $\frac{1}{2}I\omega^2 + \frac{1}{2}I(2\omega)^2 - \frac{1}{2}2I\left(\frac{3\omega}{2}\right)^2$

= $\frac{5}{2}I\omega^2 - \frac{9}{4}I\omega^2$

$\Delta K = -\frac{I\omega^2}{4}$

17. Answer (2)



$2mr_1 = mr_2$

$2r_1 = r_2$

$r_1 + r_2 = 2\ell$

$2r_1 = 2\ell - r_1$

$3r_1 = 2\ell$

$r_1 = \frac{2\ell}{3}$

$r_2 = \frac{4\ell}{3}$

$I = 2m(d_1)^2 + m(d_2)^2 \left\{\begin{matrix} d_1 = r_1\sin 30° \\ d_2 = r_2\sin 30° \end{matrix}\right\}$

$= \frac{2m\ell^2}{9} + \frac{4m\ell^2}{9}$

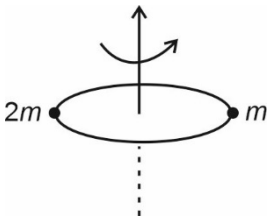
$= \frac{6m\ell^2}{9} = \frac{2m\ell^2}{3}$

18. Answer (3)

$X_{c.m.} = 0$

$Y_{c.m.} = \frac{1\times 0 + 3\times 2 + 4\times 1}{1+3+4} = \frac{5}{4}\text{m} = 1.25\text{ m}$

19. Answer (2)



By conservation of angular momentum

$3mR^2\omega = (3mR^2 + mR^2 + 2mR^2)\,\omega_1$

$\omega_1 = \frac{\omega}{2}$

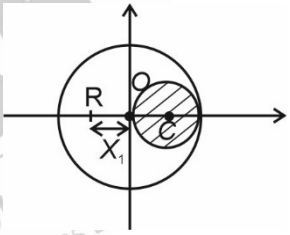
20. Answer (1)

$k_1 = \sqrt{\frac{2}{3}}\,R$

$k_2 = \sqrt{\frac{2}{5}}\,R$

$\frac{k_1}{k_2} = \sqrt{\frac{5}{3}}$

21. Answer (4)



$X_1A_1 = X_2A_2$

$X_1\left(\pi R^2 - \frac{\pi R^2}{4}\right) = \frac{R}{2}\,\frac{\pi R^2}{4}$

$3\frac{X_1}{4} = \frac{R}{8}$

$X_1 = \frac{R}{6}$ (Shift to left)

22. Answer (3)

By conservation of M.E.

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}\times\frac{2}{3}mv^2$

$mgh = \frac{5}{6}mv^2$

$v = \sqrt{\frac{6gh}{5}}$ (in both cases);

$a = \frac{g\sin\theta}{1+\frac{k^2}{R^2}}$ (depends on θ) so different.

23. Answer (3)

$$a = \frac{g\sin\theta}{1+\frac{k^2}{R^2}}$$

$$a = \frac{5}{1+\frac{2}{5}} = \frac{25}{7}\ \text{m/s}^2 \text{ (Retardation)}$$

$$0^2 = (10)^2 + 2\left(-\frac{25}{7}\right)S$$

$$100 = 2\times\frac{25}{7}\times S$$

S = 14 m

24. Answer (2)

$$\tau = I\alpha$$

$$20\times10\times10^{-2} = \frac{MR^2}{2}\alpha$$

$$2 = \frac{10\times(10)^2\times10^{-4}}{2}\alpha$$

$\alpha = 40\ \text{rad/ s}^2$

25. Answer (2)

By conservation of M.E.

$$\frac{1}{2}I\omega^2 = Mgh$$

$$\frac{1}{2}\frac{ML^2}{3}\omega^2 = Mgh$$

$$h = \frac{\omega^2L^2}{6g}$$

26. Answer (4)

$$\frac{\frac{1}{2}mv^2}{2} = \frac{1}{2}mv_1^2$$

$v_1 = \frac{v}{\sqrt{2}}$ and $f' = 2f$ and $\left(f = \frac{1}{T} = \frac{v}{2R\pi}\right)$

$$2\left(\frac{v}{2\pi R}\right) = \frac{v}{2\pi R_1\sqrt{2}}$$

$$R_1 = \frac{R}{2\sqrt{2}}$$

$L_1 = mv_1R_1 = \frac{L}{4}$

27. Answer (1)

$\vec{L} = \vec{r}\times\vec{P}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 3 & 4 & -2 \end{vmatrix}$$

$= \hat{i}(-4+4) - \hat{j}(-2+3) + \hat{k}(4-6)$

$= \vec{L} = -\hat{j} - 2\hat{k}$

Hence, angular momentum is perpendicular to x-axis.

28. Answer (2)

$$U_i = -\frac{GMm}{R}$$

$$U_f = -\frac{GMm}{3R}$$

$$W = U_f - U_i$$

$$= -\frac{GMm}{3R} + \frac{GMm}{R}$$

$$= \frac{-GMm + 3GMm}{3R} = \frac{2GMm}{3R}$$

$$W = +\frac{2mgR}{3}$$

29. Answer (2)

$$V_e = \sqrt{\frac{2GM}{R}}$$

$$V_e = \sqrt{\frac{2G\frac{4}{3}\pi R^3\rho}{R}}$$

$V_e = \sqrt{\frac{8}{3}G\pi R^2\rho} \Rightarrow V_e \propto R\sqrt{\rho}$

$$\frac{V_e}{V_p} = \frac{1}{3\sqrt{3}}$$

30. Answer (4)

$$g = \frac{GM}{R^2}$$

$$r = \frac{R}{2}$$

$$g_1 = \frac{GM}{\left(\frac{R}{2}\right)^2}$$

$g_1 = 4g$

31. Answer (1)

$$g\left(1-\frac{d}{R}\right) = g\left(1-\frac{2h}{R}\right) \Rightarrow d = 2h$$

$2 = 2h$

$h = 1$ km

32. Answer (1)

$$g = \frac{GM}{R^2}$$

$$g = \frac{G\frac{4}{3}\pi R^3\rho}{R^2} = \frac{4}{3}\pi GR\rho$$

$$\rho = \frac{3g}{4\pi RG}$$

33. Answer (3)

Acceleration due to gravity is affected by the rotation of earth.

34. Answer (1)

$$T = 2\pi\sqrt{\frac{R^3}{GM}} \text{ (Very close to planet)}$$

$$T = 2\pi\sqrt{\frac{R^3}{G\frac{4}{3}\pi R^3\rho}}$$

$$T = \sqrt{\frac{3\pi}{G\rho}}$$

35. Answer (4)

$$\text{K.E.} = \frac{1}{2}mV_0^2$$

$$= \frac{1}{2}m\left(\sqrt{\frac{GM}{2R}}\right)^2$$

$$= \frac{GMm}{4R}$$

$$\text{K.E.} = \frac{mgR}{4} \quad (GM = gR^2)$$

36. Answer (4)

Effective gravity inside the satellite is zero, therefore weight of the man inside the satellite will be zero.

37. Answer (3)

$$K = \frac{GMm}{2R}$$

$$E = -\frac{GMm}{2R}$$

$$|E| = +\frac{GMm}{2R}$$

$$\frac{K}{E} = 1 : 1$$

38. Answer (4)

$$V = -\frac{2Gm}{\frac{3a}{2}} \text{ (Point is outside the shell)}$$

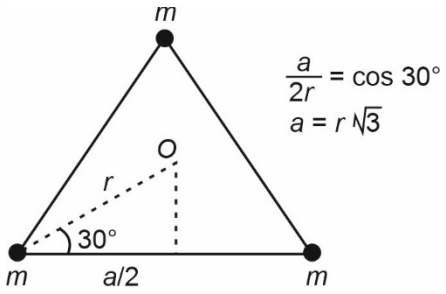
$$V = -\frac{4Gm}{3a}$$

39. Answer (2)

$$\frac{T^2}{T_1^2} = \frac{(7R)^3}{(3.5R)^3} \Rightarrow \frac{T^2}{T_1^2} = \frac{8}{1}$$

$$T_1 = \frac{T}{2\sqrt{2}} = \frac{24}{2\sqrt{2}}h = 6\sqrt{2}\,h$$

40. Answer (2)



$\frac{a}{2r} = \cos 30°$

$a = r\sqrt{3}$

$$V = -\frac{3Gm}{r}$$

$$r = \frac{a}{\sqrt{3}}$$

$$V = -\frac{3\sqrt{3}\,Gm}{a}$$

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41. Answer (4)
Total mechanical energy of the comet will be constant, as no friction is present.

42. Answer (2)
For $r < R$

$$V = -\frac{GM}{R}$$

For $r \geq R$

$$V = -\frac{GM}{r}$$

43. Answer (3)

$$E = \frac{1}{2}MV^2$$

$$V = \sqrt{\frac{2E}{M}}$$

44. Answer (4)
Gravitational force is medium independent.

45. Answer (3)
By conservation of energy

$$\frac{1}{2}mV_0^2 - \frac{GMm}{R} = -\frac{GMm}{\frac{3R}{2}}$$

$$\frac{1}{2}mV_0^2 = -\frac{2GMm}{3R} + \frac{GMm}{R}$$

$$\frac{1}{2}mV_0^2 = \frac{GMm}{3R}$$

$$V_0 = \sqrt{\frac{2GM}{3R}} \qquad (GM = gR^2)$$

$$V = \sqrt{\frac{2gR}{3}}$$

CHEMISTRY

46. Answer (3)
Work is a path function.

47. Answer (4)
w = – 30 J, q = + 10 J
$\Delta U = q + w = + 10 - 30 = - 20$ J

48. Answer (3)
At constant pressure, $\Delta H = q_P$

49. Answer (2)

$$w = -2.303\ nRT\ \log\frac{V_f}{V_i}$$

$$= -2.303 \times 1 \times 8.314 \times 300 \log\frac{1}{10}$$

= 5744 J $\approx$ 5.74 kJ

50. Answer (4)
Intensive properties do not depend on the quantity of matter e.g. molar heat capacity.

51. Answer (2)
For, $N_2(g) + O_2(g) \longrightarrow 2NO(g)$
$\Delta n_g = 0$ therefore $\Delta H = \Delta U$

52. Answer (1)
If for process $\Delta H < 0$ and $\Delta S > 0$ then ΔG is always negative therefore the process will always be spontaneous.

53. Answer (1)
$S + O_2 \longrightarrow SO_2 \quad \Delta H = - 300$ kJ
32 g sulphur gives 300 kJ heat
1g sulphur gives $\frac{300}{32}$ kJ heat
8 g sulphur gives $\frac{300 \times 8}{32}$ kJ heat
= 75 kJ heat

54. Answer (2)
$C_6H_{10} + H_2 \longrightarrow C_6H_{12}$
Heat of hydrogenation
$= \sum(\Delta_C H)_{Reactant} - \sum(\Delta_C H)_{Product}$
= (– y) + (– z) – (– x)
= x – y – z

55. Answer (2)
Given reaction is combustion of CO into CO_2.

56. Answer (2)
$CCl_4(g) \longrightarrow C(g) + 4Cl(g)$; $\Delta_r H = 4BE_{C-Cl}$
$320 = 4 \times BE_{C-Cl}$

$$BE_{C-Cl} = \frac{320}{4} = 80 \text{ kJ mol}^{-1}$$

57. Answer (3)

$\Delta H = nC_{P_{H_2O(s)}}\Delta T + \Delta_{fus}H + nC_{P_{H_2O(l)}}\Delta T$

= (1 × 35 × 10) + 6000 + (1 × 75 × 10)

= 7100 J = 7.1 kJ

58. Answer (2)

$\Delta G = \Delta H - T\Delta S = 0$

$T = \frac{\Delta H}{\Delta S} = \frac{400}{2} = 200\ K$

59. Answer (1)

$\Delta_r G° = -RT\ lnK$

$\Delta_r G° = -2.303\ RT\ log\ K$

60. Answer (4)

Increase in gaseous moles, increases the entropy.

61. Answer (3)

Third law of thermodynamics : $\lim_{T\to 0} S \to 0$

62. Answer (2)

Lattice enthalpy can be found out by the help of Born Haber cycle.

63. Answer (3)

According to Hess law

$A \longrightarrow B$	$\Delta_r H_1 = x$
$B \longrightarrow C$	$\Delta_r H_2 = y$
$\underline{C \longrightarrow D}$	$\Delta_r H_3 = z$
$A \longrightarrow D$	$\Delta_r H = \Delta_r H_1 + \Delta_r H_2 + \Delta_r H_3$
	$w = x + y + z$

64. Answer (2)

H_2SO_4 is a diprotic acid, so heat evolved = 2 × heat of neutralisation

= 2 × 13.7 = 27.4 kcal

65. Answer (2)

At equilibrium, both the opposing processes occur at the same rate.

66. Answer (3)

$2A(g) + 3B(g) \rightleftharpoons C(g) + D(g)$

$K_C = \frac{[C][D]}{[A]^2[B]^3}$

67. Answer (3)

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)\ ;\ K_C = x$

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)\ ;\ K_C = \frac{1}{x}$

$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g);\ K_C = \frac{1}{\sqrt{x}}$

68. Answer (1)

$K_P = K_C(RT)^{\Delta n_g}$, $\Delta n_g = (4 + 6) - (5 + 4) = 1$

69. Answer (3)

	A(g)	+	B(g)	⇌	C(g)
(i)	2 atm		2 atm		0
eq	(2 – x) atm		(2 – x) atm		x atm

$K_P = \frac{P_C}{P_A P_B} = \frac{x}{(2-x)(2-x)} = 1$

x = 1

At equilibrium, $P_A = P_B = P_C = 1$ atm

$P_{Total} = 3$ atm

70. Answer (4)

Higher is the value of K_C, more will be the extent of reaction.

71. Answer (2)

Increase in pressure will shift the equilibrium towards the lesser number of gaseous moles side.

72. Answer (3)

Unit of $K_P = (atm)^{\Delta n_g} = (atm)^{2-1} = atm$

73. Answer (2)

In homogeneous equilibrium, all the reactants and products are in the same phase.

74. Answer (3)

Catalyst increases the rate of forward and backward reactions that pass through the same transition state.

75. Answer (2)

Conjugate base = Acid – H^+

76. Answer (3)

$K_a = C\alpha^2 = 1 \times \left(\frac{0.001}{100}\right)^2$

$K_a = 10^{-10}$

77. Answer (4)

Salt of strong acid and strong base does not undergo salt hydrolysis.

78. Answer (3)

$$\underset{(eq.)}{} \quad \underset{-}{AB_2(s)} \rightleftharpoons \underset{(s)}{A^{2+}(aq)} + \underset{(2s)}{2B^{-}(aq)}$$

$K_{sp} = [A^{2+}] [B^{-}]^2 = (s) (2s)^2 = 4s^3$

79. Answer (2)

For acidic buffer,

$$pH = pK_a + \log \frac{[Salt]}{[Acid]}$$

$$= 4.7 + \log \frac{0.2}{0.2} = 4.7$$

80. Answer (3)

$$\underset{(eq)}{} \quad \underset{—}{AgCl(s)} \rightleftharpoons \underset{s\,M}{Ag^{+}(aq)} + \underset{(s+0.1)\,M}{Cl^{-}(aq)}$$

$K_{sp} = [Ag^{+}] [Cl^{-}] = (s) (s + 0.1)$

$1.2 \times 10^{-10} = s \times 0.1$ [$\because s + 0.1 \approx 0.1$]

$s = 1.2 \times 10^{-9}$ M

81. Answer (4)

With increase in electronegativity, acidic strength increases.

82. Answer (2)

$$pH = 7 + \frac{pK_a}{2} - \frac{pK_b}{2}$$

$$= 7 + \frac{4.7}{2} - \frac{4.7}{2} = 7$$

83. Answer (2)

For conjugate acid-base pair

$$K_a = \frac{K_w}{K_b} = \frac{10^{-14}}{2 \times 10^{-5}} = 5 \times 10^{-10}$$

84. Answer (2)

The equilibrium constant for net reaction obtained after adding two or more reactions is equal to the product of the equilibrium constants of the individual reactions.

85. Answer (4)

Reference state of carbon is graphite.

86. Answer (2)

1g water takes 2.44 kJ heat

1 mol = 18 g water takes 2.44 × 18 kJ heat

$\simeq$ 44 kJ heat

87. Answer (4)

For an isothermal expansion, $\Delta S > 0$

88. Answer (2)

Equilibrium constant depends only on temperature.

89. Answer (2)

BF_3 can accept lone pair of electrons.

90. Answer (3)

$C(s) + O_2(g) \rightarrow CO_2(g)$; $\Delta_r H = \Delta_C H(\text{carbon}) = \Delta_f H(CO_2) = -x$ kJ/mol^{-1}

BOTANY

91. Answer (2)

A tissue is a group of structurally similar or dissimilar cells that perform a common function and have a common origin.

92. Answer (1)

The plant tissues are broadly classified into two main groups namely, meristematic and permanent tissues.

93. Answer (4)

During stem elongation and leaves formation, some cells left behind from shoot apical meristem in the axil and constitute the axillary bud.

94. Answer (1)

The apical meristems produce the primary tissues of plant body and are responsible for primary growth of the plants.

95. Answer (3)

Fascicular cambium, interfascicular cambium and cork cambium are lateral meristems and these are cylindrical meristems.

96. Answer (4)

During the formation of primary plant body, specific regions of the apical meristem produce dermal tissues, ground tissues and vascular tissues. Cork is formed by cork cambium.

97. Answer (2)

Parenchyma forms the major component within plant organs.

98. Answer (2)

Both parenchyma and collenchyma cells when have chloroplast, perform photosynthesis.

99. Answer (3)

Collenchyma provides mechanical support to the young stem and petiole of leaf.

100. Answer (1)

Presence of vessels is a characteristic feature of angiosperms. Gymnosperms and pteriodophytes lack vessels in their xylem.

101. Answer (4)

In endarch type of arrangement of primary xylem, the protoxylem lies towards, centre and metaxylem lies towards the periphery.

102. Answer (3)

Cells of xylem parenchyma are thin walled and help in the radial conduction of water. These cells also store other substances like tannins.

103. Answer (3)

Protoxylem and metaxylem are primary xylem. They arise during primary growth of plant body. Metaxylem is later formed primary xylem and has vessels with broad diameter.

104. Answer (2)

At maturity, both tracheids and vessels are dead cells and are without protoplast. Tracheids are the elongated cells with tapering ends. The end walls of vessel cells are perforated plates.

105. Answer (1)

Phloem fibres (bast fibres) are sclerenchymatous fibres which are dead at maturity.

106. Answer (1)

Companion cells are closely associated with sieve tube elements. The companion cells help in maintaining the pressure gradient in the sieve tubes.

107. Answer (3)

Phloem parenchyma is living component of phloem and is absent in most of the monocotyledons.

108. Answer (3)

Subsidiary cells do not possess chloroplasts and do not carry out the photosynthesis.

109. Answer (2)

Trichomes are epidermal hairs on the stem. They are not associated with the absorption of minerals from soil or water.

110. Answer (4)

The tissues that do not come under the epidermal tissue system and vascular tissue system constitute the ground tissue system.

111. Answer (2)

Endodermis is the innermost layer of cortex. It is comprises of single layered of barrel-shaped cells.

112. Answer (2)

In dicot stems, xylem and phloem are present on the same radius and cambium is present between them. Such condition is called conjoint and open.

113. Answer (3)

The outermost layer of root is epiblema. Many of the cells of this layer protrude in the form of unicellular root hair.

114. Answer (3)

Endodermal cells are closely attached to each other and have no intercellular space between them.

115. Answer (1)

Initiation of lateral roots and vascular cambium during the secondary growth in dicot roots take place from pericycle cells.

116. Answer (2)

As compared to dicot roots, there are usually more than six xylem bundles in monocot roots.

117. Answer (4)

Starch sheath and semi-lunar patches of sclerenchyma above the phloem is found in dicot stem.

118. Answer (3)

The heartwood comprises dead elements with highly lignified walls.

119. Answer (2)

Cell theory does not state about the differentiation and specialization of cells.

120. Answer (3)

Ribosomes are found in both prokaryotic and eukaryotic cells and they provide the site for protein synthesis.

121. Answer (3)

Mycoplasma is the smallest cell. It is heterotrophic organism.

122. Answer (1)

A protoplast is a cell without cell wall.

123. Answer (2)

In prokaryotes, genetic material is basically naked, not enveloped by nuclear membrane.

124. Answer (4)

Bacteria that take up the Gram stain are Gram positive and the others that do not are called Gram negative bacteria.

125. Answer (3)

Bacterial filament is composed of three parts – filament, hook and basal body.

126. Answer (1)

Several ribosomes attach to a single mRNA form a chain called polysome.

127. Answer (2)

Peripheral proteins of the plasma membrane lie on the surface of the membrane.

128. Answer (1)

Mostly, the water molecules move across the plasma membrane from its higher concentration to the lower concentration.

129. Answer (4)

The structure in the figure represented as 'A' is smooth endoplasmic reticulum.

130. Answer (4)

Golgi apparatus remains in close association with endoplasmic reticulum as the materials to be packaged in the form of vesicles from the ER fuse with the Golgi apparatus.

131. Answer (4)

The lysosome contain enzymes for digesting carbohydrates, proteins, lipids and nucleic acids.

132. Answer (3)

Both cristae and mesosome have enzymes for respiration.

133. Answer (2)

The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins.

134. Answer (3)

p arm of chromosome is the shorter arm. In metacentric chromosome, both the arms are of equal size and in telocentric chromosome there is only one arm.

135. Answer (2)

Nucleolus is the site for active ribosomal RNA synthesis.

ZOOLOGY

136. Answer (4)

Trypsin is present in pancreatic juice

137. Answer (3)

Cholesterol and phospholipids are present in bile juice. Salivary amylase secreted by parotid salivary glands helps in digestion of starch. HCl in gastric juice is secreted by parietal cells and dipeptidases are present in intestinal juice.

138. Answer (1)

Gastric glands are present mucosa of wall of stomach.

139. Answer (4)

Insulin is a polypeptide and it is absorbed in the form of amino acids after digestion in intestine.

140. Answer (1)

Vitamins are not digested in our alimentary canal and are absorbed as such from food in intestine

141. Answer (3)

Pleurodont dentition is present in snakes. When teeth are found attached at the margin of jaw bones, the dentition is called Pleurodont.

142. Answer (4)

Gastro-oesophageal sphincter is also known as cardiac sphincter. Ileo-caecal valve prevents the back flow of undigested food from caecum into ileum.

143. Answer (2)

Micro-organisms are present in blind sac like structure called caecum which is most proximal part of large intestine.

144. Answer (3)

Muscularis of stomach contains an additional layer of oblique muscles.

145. Answer (1)

Mucosa forms irregular folds in stomach called rugae. Villi are supplied with a network of blood capillaries and a lymph capillaries called the lacteal.

146 Answer (1)

In atmospheric air pO_2 is about 158 mmHg and in alveoli it is about 104 mmHg. In atmospheric air pCO_2 is about 0.03 mmHg and in alveoli it is about 40 mmHg.

147. Answer (3)

The first step in respiration is exchange of O_2 from atmosphere with CO_2 produced within cells. This process of gaseous exchange is called breathing.

148. Answer (3)

Pinctada is a pearl-oyster, placed in phylum mollusca. Respiration through gills is called branchial respiration present in aquatic insects, molluscs, fishes and hemichordates.

149. Answer (2)

Labeo	–	Gills
Pheretima	–	Moist cuticle
Periplaneta	–	Tracheal tubes
Pteropus	–	Lungs

150. Answer (3)

Trachea divides at the level of 5th thoracic vertebra into right and left primary bronchi.

151. Answer (3)

Cartilaginous rings are present only upto initial bronchioles. Respiratory bronchioles and alveoli which are vascularised bags for gaseous exchange are not supported by cartilaginous rings.

152. Answer (1)

The part starting with the external nostrils upto the terminal bronchioles constitute the conducting part.

153. Answer (3)

Lower surface of air tight thoracic chamber is formed by diaphragm which is not included in rib cage.

154. Answer (4)

Contraction of internal intercostal muscles occur only during forceful expiration.

155. Answer (3)

Pharynx is a common passage for digestive and respiratory tract.

156. Answer (1)

Deficiency of only proteins in a child above one year of age without deficiency of calorie is called kwashiorkar. Marasmus is caused due to deficiency of both proteins and energy in the body and occurs in infants below age of one year.

157. Answer (1)

Secretion of goblet cells protect the inner lining of GIT. Parietal cells secrete HCl and Castle's intrinsic factor responsible for absorption of iron (in Fe^{+2}) and vit-B_{12} respectively.

158. Answer (4)

Peptic cells present in stomach are responsible for secretion of enzymes of gastric juice. Paneth cells are present in small intestine and secrete lysozyme to kill micro-organisms.

159. Answer (4)

Glucose and amino acids are actively absorbed in small intestine. Glycine is simplest amino acid.

160. Answer (4)

Pancreas is situated between the limbs of the 'C' shaped duodenum.

161. Answer (1)

Common bile duct and pancreatic duct open together into the duodenum as the common hepato-pancreatic duct.

162. Answer (4)

Salivary glands are present outside the buccal cavity and open in buccal cavity through their ducts. Projection on tongue surface are called papillae which contain taste-buds. Opening of hepato-pancreatic duct is guarded by sphincter of oddi.

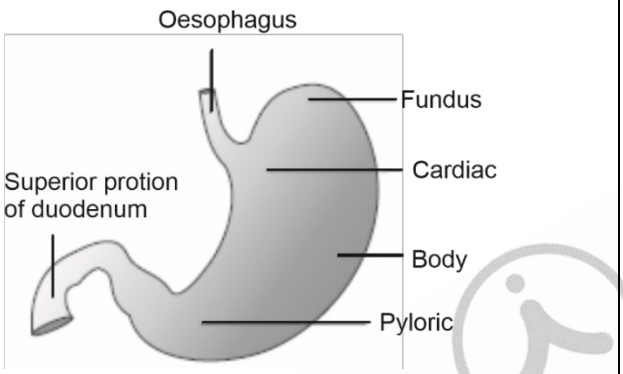
163. Answer (4)

Buccal cavity is responsible for both mechanical and chemical digestion. Only starch is hydrolysed by salivary amylase but there is no any enzyme for digestion of proteins. Disaccharides are never absorbed.

164. Answer (2)

Active forms		Inactive forms
Pepsin	–	Pepsinogen
Trypsin	–	Trypsinogen
Carboxypeptidase	–	Procarboxypeptidase

165. Answer (2)



166. Answer (4)

Carbohydrates, lipids and protein are major components of our food.

167. Answer (4)

Lactase is a disaccharidase present in succus entericus and is responsible for digestion of milk sugar lactose.

168. Answer (2)

Vomiting is the ejection of stomach contents through the mouth.

169. Answer (2)

Respiratory or exchange part of respiratory system consists of alveoli and their ducts.

170. Answer (1)

Larynx is a cartilaginous box which helps in sound production and hence called the sound box.

171. Answer (2)

Inspiration is initiated by contraction of diaphragm and external inter-costal muscles to increase volume of thoracic cavity. Intrapulmonary pressure decreases and air rushes into lungs under negative pressure.

172. Answer (3)

Ratio of TV and inspiratory reserve volume is 1 : 5 and it is equal to approximate ratio of RV and TLC.

RV : TLC = 1 : 5.

173. Answer (4)

Normal breathing rate is 12-16/min.

So air inspired/min = 500 ml × 12 or 16 ml/min

= 6000 – 8000 ml/min

174. Answer (3)

Pulmonary veins carry oxygenated blood from lungs to left auricle having minimum CO_2 and maximum O_2 in comparison to any blood vessel given in the options.

175. Answer (2)

Total volume of air inspired or expired by a normal person forcefully by his maximum effort represents vital capacity.

176. Answer (3)

Chloride ions are more within RBCs due to chloride shift at tissue site and decreases within RBCs of oxygenated blood due to reverse chloride shift at alveolar site. pH of oxygenated blood is slightly more than deoxygenated blood due to removal of CO_2 in expired air and thus decrease in H^+ concentration.

177. Answer (4)

Higher temperature favours dissociation of $Hb(O_2)$.

178. Answer (1)

Increase in pCO_2 and hydrogen ion concentration activates chemosensitive area in medulla to increase breathing rate.

179. Answer (1)

Pneumotaxic centre present in pons can moderate the function of respiratory rhythm centre situated in medulla of brain.

180. Answer (4)

Pleurisy is painful breathing due to inflammation of pleural membranes.

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